MALLA REDDY UNIVERSITY
(As perTelangana Private Universities Act No. 13 of 2020 &
G.O.Ms.No. 14, Higher Education (UE) Department)
Maisammaguda, Kompally,
Hyderabad - 500100, Telangana State.
mruniversityhyd@gmail.com
www.mallareddyuniversity.ac.in

Department of Computer Science and Engineering

R22-B.Tech-IV Year II Sem

FINAL YEAR PROJECT DEVELOPMENT THESIS TITLE SUBMISSION FORM

1. Student Details

- **Name:** B . Anjali Reddy
- **Register Number:** 2211CS010049
- **Year / Semester:** 4-2
- **Branch:** CSE
- **Section:** Alpha
- **Mobile Number:** 8309163759
- **Email ID:** 2211cs010049@mallareddyuniversity.ac.in

---

2. Project Details

- **Project Title:**
- ViraLeaf : Smarter Leaves. Healthier Crops.

- **Project Domain / Area:**
- Deep Learning, Artificial Intelligence, Image Processing , Computer Vision

- **Problem Statement:**
- Plant leaf diseases significantly affect crop yield and quality, and early detection is essential for effective prevention. Traditional manual inspection is time-consuming and often fails to identify the severity of infection in time. Existing automated systems mainly focus on disease identification without providing guidance on preventive measures. This lack of severity analysis and precautionary information makes timely disease control difficult. Therefore, an intelligent system is required to detect leaf diseases, estimate infection percentage, and suggest appropriate precautions to minimize crop damage.

- **Objective of the Project:**
  - To calculate the percentage of disease severity by analyzing the infected region of the leaf.
  - To visually highlight diseased areas to help users understand the extent of infection.
  - To provide accurate and fast disease diagnosis for early preventive action.
  - To classify healthy and diseased plant leaves using image processing and deep learning techniques

- **Problem Formulation & Literature Survey:**

Study of Existing Systems and Related Work:

Most existing plant disease detection systems use image processing and machine learning techniques such as SVM, KNN, and CNN to identify diseases from leaf images. These systems mainly focus on disease classification and provide limited information about the extent of infection.

Analysis of Techniques Used in Previous Research:

Earlier studies have applied color segmentation, texture analysis, handcrafted feature extraction, and deep learning models to improve detection accuracy. However, the output is usually restricted to disease labels without severity estimation**.**

Identification of Limitations and Research Gap:

Current systems lack accurate disease severity measurement, clear visualization of infected areas, and practical real-time usability. This highlights a research gap in integrating disease detection with percentage-based severity analysis.

Justification for the Proposed Solution:

The proposed system ViraLeaf addresses these issues by combining deep learning and image segmentation to detect plant diseases and estimate infection percentage, enabling early intervention and improved crop management**.**

3. Team Members

S.No	Name of Student	Reg . No	Contact Number	Email	Signature
1	Baireddy Anjali	2211CS010049	8309163759	baireddyanjali683@gmail.com	
2	Balaga Joshna	2211CS010051	7396319500	joshna.balaga5@gmail.com	
3	Bhavandla Shravan Kumar	2211CS010071	9908993578	shravanbhavandla@gmail.com	
4	Alti Manish Reddy	2211CS010018	9182049836	manishreddyalti@gmail.com	

4.Guide / Supervisor Details

- **Name of Guide:** Dr . Naga Chandrika Vaidya
- **Designation:** Assistant Professor
- **Department:** Computer Science and Engineering
- **Signature of Guide:**

5.Comments / Remarks :

Dean-CSE
Signature: